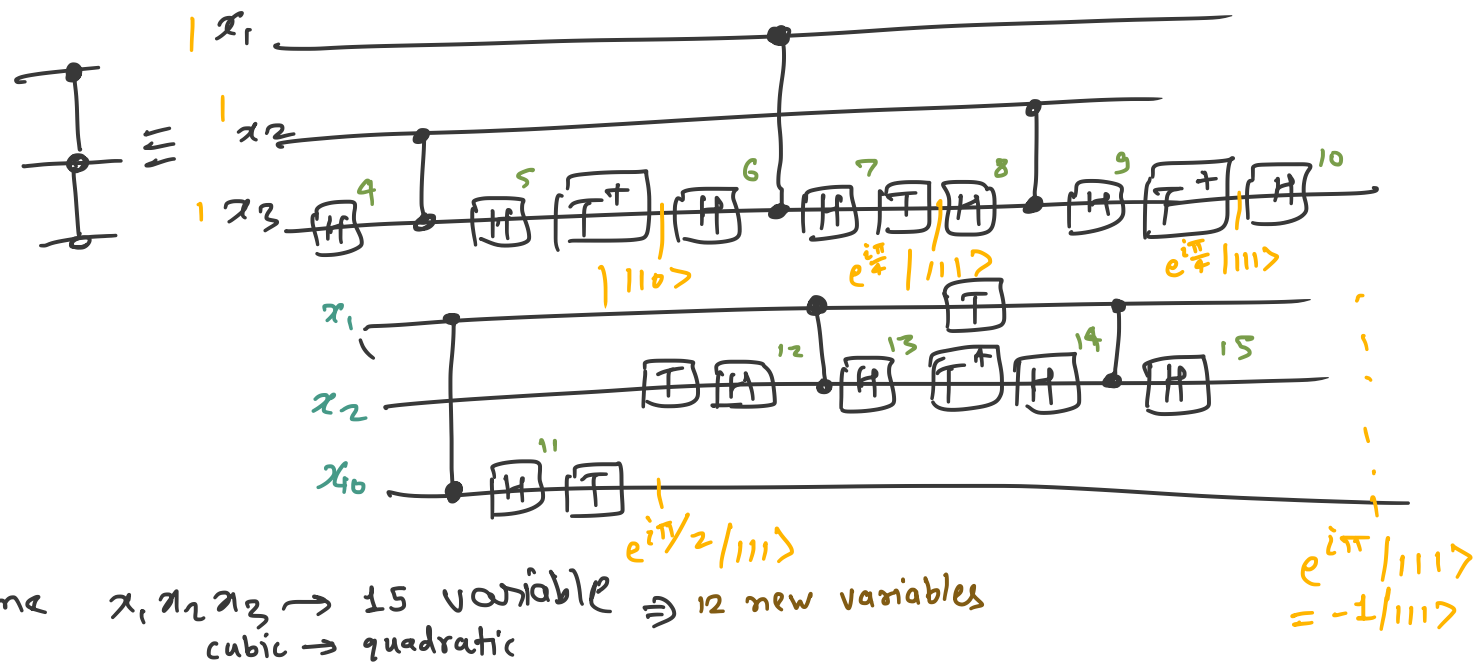
CCZ Decomposition –



one $x_1 x_2 x_3 \rightarrow$ 15 variable $\Rightarrow$ 12 new variables
cubic $\rightarrow$ quadratic

$$C(x) = 4x_3x_4 + 4x_2x_3 + 4x_4x_5 + 7x_5 + 4x_5x_6 + 4x_1x_6$$
(in Z_8)
$$+ 4x_6x_7 + x_7 + 4x_7x_8 + 4x_2x_8 + 4x_8x_9 + 7x_9 + 4x_9x_{10}$$
$$+ 4x_1x_{10} + 4x_{10}x_{11} + x_{11} + x_2 + 4x_2x_{12} + 4x_1x_{12}$$
$$+ 4x_{12}x_{13} + (7x_{13} + x_1) + 4x_{13}x_{14} + 4x_1x_{14} + 4x_{14}x_{15}$$

Simon's Algo (like)

$n=4,\ m=1$ but in Simon's $m \geq n$

let $f(x) = x_2 + x_3 + x_2x_3 + x_1x_2x_3 + x_2x_3x_4$ (oracle)

$\Rightarrow$ does this have secret shift s (or a)
such that $f(x+a) = f(x)\ \forall x$.

$$f(x) = x_2 + x_3(1+x_2) + x_2x_3(x_1+x_4)$$

	x_1	x_2	x_3	x_4	$F(x)$
$\Rightarrow$ 0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	1
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	0
10	1	0	1	0	1
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

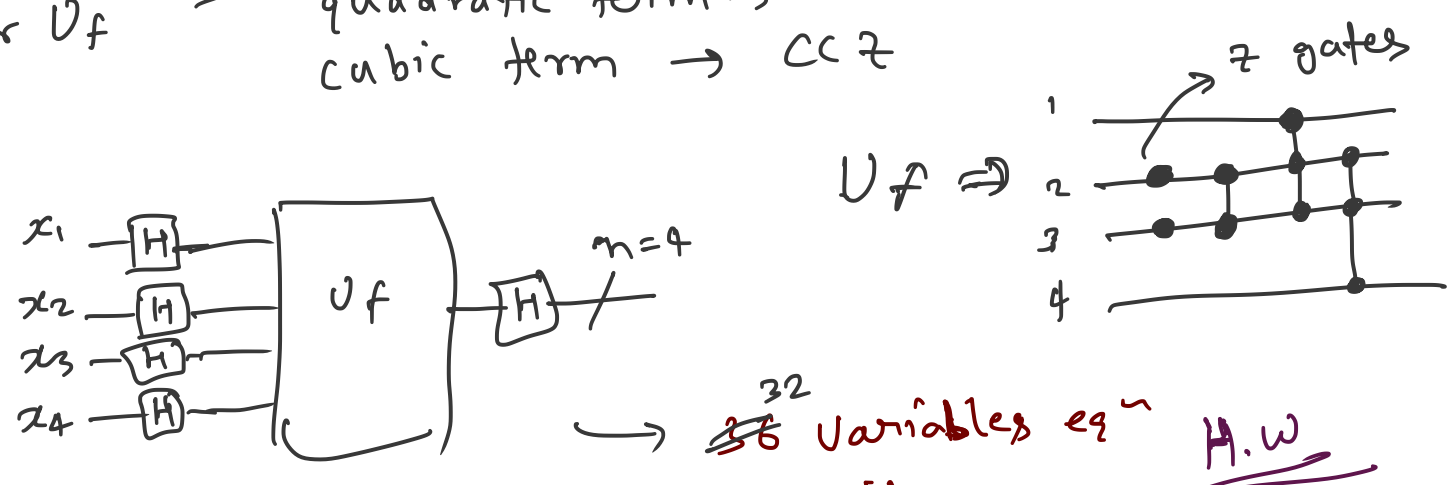
here x_2 & x_3 are interchangeable
and x_1 & x_4 are also interch
↳ for faster evaluation of $f(x)$
{ might need to figure out in code }

how to find a –
first count 0's & 1's – 6 0's
and 10 1's, even so 'a' possible.
→ take $x=0$ (or any value) and
for all x' s.t. $f(x') = f(x)$
check - let $a = x \oplus x'$
is $f(x \oplus a) = f(x)$? $\forall x$
if it matches for all x then we
found a.
$\Rightarrow$ since, $x=0$, $a = x'$
eg. $x' = 1001 \rightarrow a = 1001$
$x=1$ $f(0001 \oplus 1001) = f(1000) = f(0001) = 0$
$x=4$ $f(0100 \oplus 1001) = f(1101) = f(0100) = 1$

$\Rightarrow$ Other method to find a –
by observation – put $x_1 = 1 + x_1$
and $x_4 = 1 + x_4$
$f(x)$ does not change
and therefore $a = 1001$

Eqn to circuit –

circuit for U_f $\Rightarrow$
linear term $\rightarrow$ Z
quadratic term $\rightarrow$ CZ
cubic term $\rightarrow$ CCZ



$\rightarrow$ ~~36~~ 32 variables eqn H.W
make it

$$32 = \underset{H}{4} + \underset{CCZ}{12} + \underset{CCZ}{12} + \underset{H}{4}$$

max terms possible in 36 variables quadratic eqn $\Rightarrow {}^nC_1 + {}^nC_2 = 36 + 18 \times 35$

Total terms here $\Rightarrow \underset{H}{4} + (\underset{Z}{2} + \underset{CZ}{1} + \underset{CCZ}{25} + \underset{CCZ}{25}) + 4 = 61$

↳ some terms could be cancelled bcz of consecutive Hs but lets not do it right now.